Problem 6 – Siege

Problem

Does a 3-manifold obtained by Dehn surgery on an L-space knot have at most two prime summands?

Result

Throughout, an *L-space knot* is a knot $K \subset S^3$ such that $S^3_r(K)$ is an L-space for some slope $r \in \mathbb{Q}$. The answer is **yes**, for every slope.

Main Theorem. *Let $K \subset S^3$ be an L-space knot and $r \in \mathbb{Q} \cup \{\infty\}$. Then $S^3_r(K)$ has at most two prime summands.*

The bound is sharp. By Moser [13], $S^3_6(T(2,3)) \cong L(2,1)\#L(3,2)$, which has two prime summands. More generally, if a cable $C_{a,b}(J)$ (with winding number a) is an L-space knot, then its surgery at the cabling slope ab is $L(a,b)\#S^3_{b/a}(J)$ [3].

Overview of the proof

Suppose $Y = S^3_r(K)$ has at least three prime summands. Classical results (Gabai; Gordon–Luecke) force $r = p$ to be a nonzero integer and K to be nontrivial. Since $\pi_1(Y)$ is normally generated by one element, Howie's solution of the Scott–Wiegold conjecture shows that at most two summands have nontrivial first homology; hence some summand B is an integer homology sphere other than S^3. A theorem of Gordon–Luecke supplies a lens space summand L of order $a \geq 2$.

Two independent arguments then collide. *Topologically*, K is fibred (Ni, Ghiggini), and in the $|p|$-fold cyclic cover of Y the summand B lifts to $|p|$ copies, while the fibred structure shows that the cover has fundamental group generated by $2g$ elements. Grushko's theorem then gives $|p| \leq g$. *Floer-theoretically*, the lens space summand forces $HF^+(S^3_p(K), i)$ to be invariant, up to grading shift, under $i \mapsto i + |p|/a$ in the labelling of Spin^c structures by $\mathbb{Z}/|p|$. Using the Ozsváth–Szabó integer surgery formula we compute, for $2 \leq |p| \leq 2g-2$, a translation-invariant multiset τ of each such group: its size depends only on how many integers $s \equiv i \pmod{|p|}$ lie in $[-(g-1), g-1]$, and its elements depend only on these integers. The size count forces $|p| \mid 2g-1$, and a symmetry argument comparing the class 0 (whose τ has repeated entries) with any other class in the same orbit (whose τ has distinct entries) rules out $|p| \mid 2g-1$, $|p| < 2g-1$. Since $|p| \leq g$, this is a contradiction.

The cover bound (Proposition 4.1), the computation of τ (Proposition 5.3) and the arguments of Sections 2–6 are argued here; the inputs we use from the literature are collected, with references, in Section 1.

1 Conventions and cited results

We write $\mathbb{F} = \mathbb{Z}/2$. For a knot $K \subset S^3$, M_K is its exterior, μ and λ are its meridian and Seifert longitude on $T = \partial M_K$, and $g = g(K)$ is its genus. Prime decompositions are those of Kneser and Milnor [9, 11]: every closed oriented 3-manifold $Y \neq S^3$ is a connected sum $P_1 \# \cdots \# P_k$ of prime manifolds $P_i \neq S^3$, and k and the P_i are unique (up to order and homeomorphism); k is the number of prime summands, and S^3 has none. A closed manifold with at least two prime summands is reducible, since a splitting sphere does not bound a ball.

Mirroring. If mK denotes the mirror of K, then $S^3_r(mK) \cong -S^3_{-r}(K)$, and reversing orientation does not change the number of prime summands. An L-space surgery is never at slope 0 (it has $b_1 = 1$). If K has an L-space surgery at a negative slope, then mK has one at a positive slope. Hence we may assume that K is a *positive L-space knot*, meaning that it has an L-space surgery at some positive slope. The surgery slope p studied below is then still an arbitrary integer of either sign, and both signs are treated.

We use the following known results.

Cited Theorem A (Gabai [1])**.** *If $K \subset S^3$ is a nontrivial knot, then $S^3_0(K)$ is irreducible.*

Cited Theorem B (Gordon–Luecke [4])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then r is an integer.*

Cited Theorem C (Howie [8]; the Scott–Wiegold conjecture)**.** *A free product of three nontrivial cyclic groups is not the normal closure of a single element.*

Cited Theorem D (Gordon–Luecke [5])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then one of its prime summands is a lens space (other than S^3 and $S^1 \times S^2$).*

We use only the weak consequence that some prime summand is an L-space L with $|H_1(L)| \geq 2$ (lens spaces are L-spaces [18]).

Cited Theorem E (Ni [15]; Ghiggini [2] for $g = 1$; Ozsváth–Szabó [18])**.** *An L-space knot K is fibred, and its fibre is a minimal genus Seifert surface F, of genus g, with one boundary component.*

(Ozsváth and Szabó [18] showed that the knot Floer homology of an L-space knot in its top Alexander grading is $\mathbb{F}$; Ni and Ghiggini showed that this characterises fibred knots.)

Cited Theorem F (Grushko [6], Neumann [14]; see also [10])**.** *If G_1, G_2 are finitely generated groups, then* $\operatorname{rank}(G_1 * G_2) = \operatorname{rank} G_1 + \operatorname{rank} G_2$*, where* rank *is the minimal number of generators.*

Together with van Kampen's theorem, $\pi_1(M_1 \# M_2) \cong \pi_1(M_1) * \pi_1(M_2)$, this gives $\operatorname{rank} \pi_1(M_1 \# M_2) = \operatorname{rank} \pi_1(M_1) + \operatorname{rank} \pi_1(M_2)$.

Cited Theorem G (Perelman [20, 21, 22]; see [12])**.** *A closed simply connected 3-manifold is homeomorphic to S^3.*

This can be avoided; see Remark 4.5.

Heegaard Floer input. An *L-space* is a rational homology sphere Y with $HF^+(Y, \mathfrak{t}) \cong \mathcal{T}^+ := \mathbb{F}[U, U^{-1}]/U \cdot \mathbb{F}[U]$ for every $\mathfrak{t} \in \operatorname{Spin}^c(Y)$ [18]. Let K be a positive L-space knot of genus $g \geq 1$ and $p \neq 0$ an integer.

(H1) *The integers V_s* ([19, 18]; the notation is that of Ni–Wu [16]). There are integers V_s, $s \in \mathbb{Z}$, with

$$V_s \geq 0, \qquad V_s - V_{s+1} \in \{0, 1\}, \qquad V_s = 0 \iff s \geq g, \qquad V_{-s} = V_s + s.$$

Put $H_s := V_{-s}$; thus $H_s = V_s + s$, and $H_s = 0 \iff s \leq -g$.

(H2) *The integer surgery mapping cone, plus version* (Ozsváth–Szabó [19]). The set $\mathrm{Spin}^c(S^3_p(K))$ is labelled by $\mathbb{Z}/|p|$. For each label i, $HF^+(S^3_p(K), i)$ is isomorphic, as a relatively $\mathbb{Z}$-graded $\mathbb{F}[U]$-module, to the homology of the mapping cone of

$$D = \sum_s (v_s + h_s) \colon \bigoplus_{s \equiv i} A^+_s \longrightarrow \bigoplus_{s \equiv i} B^+_s$$

(direct sums over $s \equiv i \pmod{|p|}$), where $v_s \colon A^+_s \to B^+_s$ and $h_s \colon A^+_s \to B^+_{s+p}$ are homogeneous chain maps. For a positive L-space knot, $H_*(A^+_s) \cong H_*(B^+_s) \cong \mathcal{T}^+$, and on homology v_s is U^{V_s} and h_s is U^{H_s} [18, 19].

Why homology-level maps suffice. $\mathbb{F}[U]$ is a graded principal ideal domain, hence hereditary. So every complex of graded $\mathbb{F}[U]$-modules is quasi-isomorphic to its homology (it is formal), and in the derived category $\mathrm{Hom}(X, Z) = \mathrm{Hom}(H(X), H(Z)) \oplus \mathrm{Ext}^1(H(X), H(Z))$, with the appropriate degree shift. The target homology $\bigoplus \mathcal{T}^+$ is injective: $\mathcal{T}^+$ is divisible, $\mathbb{F}[U]$ is Noetherian, and over a Noetherian ring direct sums of injective modules are injective. So the Ext^1 term vanishes, and the morphism D is determined up to isomorphism in the derived category by its homology map. The mapping cone of a morphism is determined up to isomorphism by the morphism. Hence the homology of the cone, with its relative grading, can be computed from the homology-level maps U^{V_s} and U^{H_s}.

(H3) *Affine labelling* [19]. The labelling $\mathrm{Spin}^c(S^3_p(K)) \leftrightarrow \mathbb{Z}/|p|$ is affine over $H_1(S^3_p(K)) = \mathbb{Z}/|p| = \langle [\mu] \rangle$, and $[\mu]$ acts by $i \mapsto i + 1$. (The sign convention does not matter below.)

(H4) *Connected sums* (Ozsváth–Szabó [17]). $\mathrm{Spin}^c(Y_1 \# Y_2) = \mathrm{Spin}^c(Y_1) \times \mathrm{Spin}^c(Y_2)$, equivariantly for $H_1(Y_1) \oplus H_1(Y_2)$, and $CF^-(Y_1 \# Y_2, (\mathfrak{t}_1, \mathfrak{t}_2)) \simeq CF^-(Y_1, \mathfrak{t}_1) \otimes_{\mathbb{F}[U]} CF^-(Y_2, \mathfrak{t}_2)$. If L is an L-space then $CF^-(L, \mathfrak{t}) \simeq \mathbb{F}[U]$, so $HF^+(L \# Y'', (\mathfrak{t}, \mathfrak{t}'')) \cong HF^+(Y'', \mathfrak{t}'')$ as relatively graded $\mathbb{F}[U]$-modules, up to a shift. Lens spaces are L-spaces [18].

2 Reduction

Suppose $S^3_r(K)$ has at least three prime summands; we derive a contradiction.

If K is the unknot, its surgeries are S^3, $S^1 \times S^2$ and lens spaces, each with at most one prime summand. So K is nontrivial. Slope ∞ gives S^3, and slope 0 gives an irreducible manifold by Theorem A. By Theorem B, a non-integral slope also gives an irreducible manifold. An irreducible closed 3-manifold has at most one prime summand.

So $r = p$ is a nonzero integer. After mirroring (Section 1), K is a nontrivial positive L-space knot of genus $g \geq 1$, and $p \neq 0$ has either sign. Write

$$Y = S^3_p(K) = P_1 \# \cdots \# P_k, \qquad P_i \text{ prime}, \ k \geq 3.$$

$H_1(Y) = \mathbb{Z}/|p|$ is finite. So no P_i is $S^1 \times S^2$, and each $H_1(P_i)$ is a direct summand of $\mathbb{Z}/|p|$, hence finite cyclic.

3 Two topological consequences

Lemma 3.1 (a homology sphere summand)**.** *Some P_i is an integer homology sphere $B \neq S^3$.*

Proof. $\pi_1(M_K)$ is normally generated by μ, and $\pi_1(Y)$ is a quotient of it, so $\pi_1(Y)$ is normally generated by one element. Suppose three summands, say P_1, P_2, P_3, had nontrivial H_1. Then $\pi_1(Y) = \pi_1(P_1) * \cdots * \pi_1(P_k)$ maps onto $\pi_1(P_1) * \pi_1(P_2) * \pi_1(P_3)$, by killing the other factors, and then onto $H_1(P_1) * H_1(P_2) * H_1(P_3)$. This target is a free product of three nontrivial cyclic groups, and it is normally generated by the image of one element. This contradicts Theorem C. So at most two summands have nontrivial H_1. Since $k \geq 3$, some P_i has $H_1(P_i) = 0$, and $P_i \neq S^3$ because it is a prime summand. □

Write $Y \cong A \# B$, where B is this homology sphere and A is the connected sum of the other summands.

Lemma 3.2 (an L-space summand)**.** *$Y \cong L \# Y''$, where L is a lens space with $a := |H_1(L)| \geq 2$. Hence a divides $|p|$ and $|p| \geq 2$.*

Proof. Y is reducible since $k \geq 2$. By Theorem D, some prime summand is a lens space $L \neq S^3, S^1 \times S^2$, so $a \geq 2$. Also $H_1(L)$ is a direct summand of $\mathbb{Z}/|p|$. □

4 The cover bound

Proposition 4.1 (cover bound)**.** *Let $K \subset S^3$ be fibred, with fibre F of genus g and one boundary component, and let $p \neq 0$ be an integer. If $S^3_p(K) \cong A \# B$ with B an integer homology sphere other than S^3, then, with $N := |p|$ and $\tilde{A}$ the connected N-fold cyclic cover of A defined below,*

$$|p| \cdot \operatorname{rank} \pi_1(B) + \operatorname{rank} \pi_1(\tilde{A}) \leq 2g, \qquad \text{and hence} \qquad |p| \leq g.$$

Setup. $Y = S^3_p(K) = M_K \cup_T V$, where the meridian disc of the solid torus V is bounded by $\alpha = p\mu + \lambda$ on $T = \partial M_K$. $H_1(M_K) = \mathbb{Z}\langle\mu\rangle$; the class of a loop is its algebraic intersection number with F, and $\lambda = 0$ in $H_1(M_K)$. So $H_1(Y) = \mathbb{Z}/N$. Let $\varphi\colon \pi_1(Y) \to H_1(Y) = \mathbb{Z}/N$ be the abelianisation, which is onto, and let $q\colon Y_N \to Y$ be the connected N-fold cyclic cover with group $\ker \varphi$.

Lemma 4.2 (lower bound)**.** *$Y_N \cong \tilde{A} \# B \# \cdots \# B$ with N copies of B, and* $\operatorname{rank} \pi_1(Y_N) = \operatorname{rank} \pi_1(\tilde{A}) + N \operatorname{rank} \pi_1(B)$ *with* $\operatorname{rank} \pi_1(B) \geq 2$.

Proof. 1. Let S be the splitting sphere, so $Y = A^\circ \cup_S B^\circ$, where $^\circ$ means that one open ball is removed.

2. $\pi_1(Y) = \pi_1(A) * \pi_1(B)$, and $H_1(A) = H_1(Y) \cong \mathbb{Z}/N$ because $H_1(B) = 0$.

3. The restriction of φ to $\pi_1(B^\circ) = \pi_1(B)$ factors through $H_1(B) = 0$. So $q^{-1}(B^\circ)$ is N disjoint copies of B°, each mapped homeomorphically onto B°.

4. S is simply connected, so $q^{-1}(S)$ is N spheres.

5. The restriction of φ to $\pi_1(A^\circ) = \pi_1(A)$ is the surjection $\pi_1(A) \to H_1(A) \cong \mathbb{Z}/N$. So $q^{-1}(A^\circ)$ is connected: it is the induced N-fold cyclic cover $\tilde{A}$ of A with N balls removed (the removed ball lifts to N balls).

6. Therefore Y_N is $\tilde{A}$ minus N balls, with a copy of B° glued to each boundary sphere. The gluing graph is a star, hence a tree, so $Y_N \cong \tilde{A}\#B\#\cdots\#B$ with N copies of B. (If $N = 1$ the cover is trivial and $\tilde{A} = A$.)

7. By van Kampen and Theorem F, $\operatorname{rank}\pi_1(Y_N) = \operatorname{rank}\pi_1(\tilde{A}) + N\operatorname{rank}\pi_1(B)$.

8. $\pi_1(B)$ is perfect, since $H_1(B) = 0$, and nontrivial, by Theorem G, since $B \neq S^3$. A nontrivial cyclic group has nontrivial abelianisation, so $\pi_1(B)$ is not cyclic, and $\operatorname{rank}\pi_1(B) \geq 2$. □

Lemma 4.3 (upper bound)**.** $\operatorname{rank}\pi_1(Y_N) \leq 2g$.

Proof. 1. *The exterior lifts to a mapping torus.* On $\pi_1(M_K)$, φ is the composite $\pi_1(M_K) \to H_1(M_K) = \mathbb{Z} \to \mathbb{Z}/N$ with $\mu \mapsto 1$. So $E := q^{-1}(M_K)$ is connected and is the N-fold cyclic cover dual to F. It is the mapping torus of h^N, where h is the monodromy, and it fibres over S^1 with fibre $\tilde{F}$, a homeomorphic lift of F. So $\pi_1(E) = \pi_1(\tilde{F}) \rtimes \langle t\rangle$, and the fibration map $\pi\colon \pi_1(E) \to \mathbb{Z}$ has kernel $\pi_1(\tilde{F})$ and sends $t \mapsto 1$.

2. *The boundary lifts to one torus.* $\mu \in \pi_1(T)$ maps onto a generator of $\mathbb{Z}/N$. So $\tilde{T} := q^{-1}(T)$ is one torus, and $q|\colon \tilde{T} \to T$ has degree N.

3. *The solid torus lifts to one solid torus.* μ meets α once, so μ is isotopic in V to the core of V. Hence $\pi_1(V)$ maps onto $\mathbb{Z}/N$, and $\tilde{V} := q^{-1}(V)$ is one solid torus with $\partial\tilde{V} = \tilde{T}$.

4. *The filling curve.* The meridian disc of V lifts to N discs. Let $\beta \subset \tilde{T}$ be the boundary of one of them. Then q maps β homeomorphically onto α, so $q_*[\beta] = [\alpha]$, and $Y_N = E \cup \tilde{V}$ is the Dehn filling of E along β.

5. *Intersection numbers.* For a degree-N covering $f\colon \tilde{T} \to T$ of tori, $f_*x \cdot f_*y = N(x \cdot y)$ for $x, y \in H_1(\tilde{T})$. This follows from the transfer $f^!$: $f_*x \cdot z = x \cdot f^!z$ and $f^!f_* = N$, since deck transformations of a torus act trivially on H_1. Now $\tilde{F} \cap \tilde{T} = \partial\tilde{F}$ is a single curve $\tilde{\lambda}$ with $q_*[\tilde{\lambda}] = [\lambda]$. Hence
$$N(\beta \cdot \tilde{\lambda}) = \alpha \cdot \lambda = (p\mu + \lambda) \cdot \lambda = \pm p,$$
so $\beta \cdot \tilde{\lambda} = \pm 1$. This holds for either sign of p.

6. *β goes once around.* On $\pi_1(\tilde{T})$, the map π is algebraic intersection with the fibre boundary $\tilde{\lambda}$. So $\pi(\beta) = \pm 1$, and β represents $t^{\pm 1}w$ for some $w \in \pi_1(\tilde{F})$, for any choice of basepoint path.

7. *Conclusion.* By van Kampen, $\pi_1(Y_N) = \pi_1(E)/\langle\langle\beta\rangle\rangle$. In this quotient $t = w^{\mp 1}$ lies in the image of $\pi_1(\tilde{F})$. So $\pi_1(Y_N)$ is generated by the image of $\pi_1(\tilde{F})$, which is free of rank $2g$, and $\operatorname{rank}\pi_1(Y_N) \leq 2g$. □

Proof of Proposition 4.1. Combining Lemmas 4.2 and 4.3: $2N \leq N\operatorname{rank}\pi_1(B) \leq N\operatorname{rank}\pi_1(B) + \operatorname{rank}\pi_1(\tilde{A}) = \operatorname{rank}\pi_1(Y_N) \leq 2g$, so $|p| = N \leq g$. □

Remark 4.4. The bound is attained: for $T(2,3)$ and $p = 1$, Y is the Poincaré sphere and $2 = 2$. For a nontrivial fibred knot J and $p \geq 2$, the cable $C_{p,1}(J)$ is fibred of genus $p\,g(J) \geq p$, and its p-surgery is $L(p,1)\#S^3_{1/p}(J)$ [3], in agreement with the bound. (This cable is not an L-space knot; Proposition 4.1 needs only fibredness.)

Remark 4.5 (avoiding Theorem G)**.** Replace rank by Heegaard genus h. Since $\beta \cdot \tilde{\lambda} = \pm 1$, Y_N is the closed manifold of an open book with page $\tilde{F}$, so $h(Y_N) \leq 2g$. By Haken's lemma [7], $h(Y_N) = h(\tilde{A}) + Nh(B)$. Finally $h(B) \geq 2$, since the closed 3-manifolds of Heegaard genus at most 1 are S^3, $S^1 \times S^2$ and lens spaces. This again gives $N \leq g$.

5 Heegaard Floer input

In this section K is a nontrivial positive L-space knot of genus g, and $p \neq 0$ is an integer with $N := |p|$.

5.1 Coset invariance

Suppose $S^3_p(K) \cong L \# Y''$, where L is an L-space with $a = |H_1(L)| \geq 2$ and Y'' is a closed oriented 3-manifold. If the homeomorphism reverses orientation, replace (L, Y'') by $(-L, -Y'')$; $-L$ is again an L-space of the same order. Put $b := N/a$. Then b is an integer and $b \not\equiv 0 \pmod N$.

Lemma 5.1 (coset invariance)**.** *For every label i, $HF^+(S^3_p(K), i) \cong HF^+(S^3_p(K), i+b)$ as relatively graded $\mathbb{F}[U]$-modules, up to a degree shift.*

Proof. $H_1(L)$ is a subgroup of order a of $\mathbb{Z}/N = \langle [\mu] \rangle$, so it is $\langle b[\mu] \rangle$. By (H4), the Spin^c structures with a fixed Y''-component $\mathfrak{t}''$ form one $H_1(L)$-orbit, and all of them have $HF^+ \cong HF^+(Y'', \mathfrak{t}'')$ up to shift. By (H3), the $H_1(L)$-orbits are exactly the cosets $i + b\mathbb{Z}$ of labels. □

5.2 The invariant τ

Definition 5.2. Let M be a graded $\mathbb{F}[U]$-module that is finite-dimensional in each degree. Record degrees by *level*, meaning half the grading, so that U lowers the level by 1. Define $\tau(M)$ to be the multiset of levels k, where k appears with multiplicity $\dim(M_k / U \cdot M_{k+1})$.

τ is an isomorphism invariant, and shifting the grading translates τ. If $M = \mathcal{T}^+ \oplus \bigoplus_l \mathbb{F}[U]/U^{e_l}$, then $\tau(M)$ is the multiset of the top levels of the finite summands, because U maps onto $\mathcal{T}^+$; in particular $|\tau(M)|$ is the number of finite summands. Clearly $\tau(M \oplus M') = \tau(M) \cup \tau(M')$.

5.3 The level model and the structure of each class

Fix a class $c \in \mathbb{Z}/N$. Let $s_1 < s_2 < \cdots < s_m$ be the integers $s \equiv c \pmod N$ with $|s| \leq g - 1$. Consecutive ones differ by N, and

$$m = n_c := \#\{s \equiv c \pmod N : \ |s| \leq g - 1\}.$$

Level model. By (H2) we may compute with the homology-level maps U^{V_s} and U^{H_s} between copies of $\mathcal{T}^+$.

- Record each tower $\mathcal{T}^+$ by its *bottom level*.
- Regrade each B-tower so that the cone maps preserve levels. This is possible: in one class the A's and B's form a connected graph (each B is joined to its at most two adjacent A's), and the relative grading of the cone is homogeneous. A map U^e from a tower with bottom x then lands in a tower with bottom $x + e$.
- Under this regrading, an A-element at level k maps to the B-element at level k in each adjacent B that is *present at level k*, meaning that its bottom is $\leq k$. All exponents are ≥ 0, so every present B has its adjacent A's present.
- So at level k the cone map is the coboundary $\delta_k \colon \mathbb{F}^{\text{vertices}} \to \mathbb{F}^{\text{edges}}$ of a graph G_k. Its vertices are the A's with bottom $\leq k$, and its edges are the B's with bottom $\leq k$, each B joining its adjacent A's.

- U acts on cochains by restriction from G_{k+1} to G_k.
- The cone has two terms and D commutes with U, so $H(\text{cone}) = \ker D \oplus \operatorname{coker} D$ as graded $\mathbb{F}[U]$-modules.

The true relative grading is 2·level+constant on each side, because both satisfy the same constraints on a connected graph. So τ, computed in levels, is the invariant τ above up to translation.

Proposition 5.3 (structure of a class)**.** *Let c and $s_1 < \cdots < s_m$ be as above.*

(A) Let $0 < p \le 2g-2$ and $m \ge 1$. Then $\tau(HF^+(S^3_p(K), c)) = \{y - 1 : \ y \in Y_c\}$, where Y_c is the multiset of the $m-1$ interior peak *levels*

$$y_j = \frac{\mu_j^2 - p^2/4}{2p}, \qquad \mu_j = s_j + \frac{p}{2} \qquad (1 \le j \le m-1).$$

In particular $|\tau| = m - 1$.

(B) Let $p = -N < 0$ and $m \ge 1$. Let Π_c be the multiset of the $m+1$ peak *levels*

$$b(t) = -\frac{u^2 - N^2/4}{2N}, \qquad u = t + \frac{N}{2}, \qquad t \in \{s_1 - N,\, s_1,\, s_2, \ldots, s_m\},$$

and let M be its maximum. Then $\tau(HF^+(S^3_p(K), c))$ is $\{y - 1 : \ y \in \Pi_c\}$ with one copy of $M - 1$ removed. In particular $|\tau| = m$.

Neither multiset depends on K.

Proof of (A). In class c the A's and B's form one infinite path $\cdots A_{s-p} - B_s - A_s - B_{s+p} - \cdots$. Writing $x(s)$ for the bottom of A_s, the bottoms must satisfy $x(s+p) - x(s) = H_s - V_{s+p}$, since B_{s+p} receives U^{H_s} from A_s and $U^{V_{s+p}}$ from A_{s+p}. One solution is

$$x(s) = \frac{s(s-p)}{2p} - V_s;$$

indeed $x(s+p) - x(s) = s - V_{s+p} + V_s = H_s - V_{s+p}$, using $H_s = V_s + s$. The B between s and $s+p$ has bottom

$$y = x(s) + H_s = \frac{s(s+p)}{2p} = \frac{\mu^2 - p^2/4}{2p}, \qquad \mu = s + \frac{p}{2}.$$

We have $x(s) \to +\infty$ as $|s| \to \infty$: for $s \ge g$, $V_s = 0$, and for $s < 0$, $V_s = V_{-s} + |s| \le g + |s|$. So the quadratic term dominates, and each G_k is a finite subforest of a path.

The coboundary of a finite forest is onto. So D is surjective and $HF^+ \cong \ker D$. At level k, $\ker \delta_k$ is the space of functions on the components of G_k, and U is restriction of functions. Hence $\dim(M_k/U \cdot M_{k+1})$ is the number of *merges*: the number of components of G_k minus the number of components of G_{k+1} that contain a vertex of G_k.

Edges with a zero exponent. $V_s = 0$ iff $s \ge g$, and $H_s = 0$ iff $s \le -g$. A B with both exponents 0 would be B_{s+p} with $s \le -g$ and $s + p \ge g$, so $p \ge 2g$, which is excluded. A B with exactly one exponent 0 appears at the same level as its endpoint A_e on that side. That endpoint is new at that level. Its other edge has a positive exponent at A_e, since $V_e = H_e = 0$ would need $e \ge g$ and $e \le -g$; so that other edge appears strictly later. The far endpoint of B is old, since B's exponent there is positive. So at this level B attaches the new vertex A_e to one old component, and merges no old components.

Edges with both exponents ≥ 1. The edge B_{s+p} between A_s and A_{s+p} has exponents $H_s \geq 1$ (iff $s \geq -(g-1)$) and $V_{s+p} \geq 1$ (iff $s+p \leq g-1$). So these are exactly the $m-1$ edges between consecutive s_j, s_{j+1}: the interior peaks. Each appears strictly after both of its endpoints, and since G is a path, each one joins two distinct components of the old graph.

So the merges from level k to level $k+1$ are the interior peaks at level $k+1$. Hence $\tau = \{y-1 : y \text{ an interior peak}\}$ and $|\tau| = m-1$. □

Proof of (B). Here $h_s \colon A_s \to B_{s-N}$. Let C be the class-c cone.

Truncation, step 1. Let $P = \{A_s : s < g\} \cup \{B_t : t < g\}$. It is a subcomplex: for $s < g$, A_s maps to B_s and B_{s-N}, and both indices are $< g$. The quotient $Q = C/P = \{A_s, B_s : s \geq g\}$ has differential $D_Q = v_s + h_s$, where $v_s \colon A_s \to B_s$ is the identity since $V_s = 0$, and h_s lowers the index by N and is present only while $s - N \geq g$. On finitely supported chains, the identity plus a part that lowers the index, with indices bounded below, is bijective: invert it by a finite Neumann series. So Q is acyclic and $H(C) \cong H(P)$.

Truncation, step 2. Let $P' = \{A_s : |s| < g\} \cup \{B_t : -g-N < t < g\}$. It is a subcomplex of P: for $|s| < g$ we have $s - N > -g - N$. The quotient $R_0 = P/P' = \{A_s : s \leq -g\} \cup \{B_t : t \leq -g-N\}$ has differential $h_s + v_s$, where $h_s \colon A_s \to B_{s-N}$ is the identity since $H_s = 0$, and $v_s \colon A_s \to B_s$ is present only while $s \leq -g - N$. Relative to the pairing $A_s \leftrightarrow B_{s-N}$, the second part raises the B-index by N, and indices are bounded above. So it is again identity plus nilpotent on finitely supported chains, and R_0 is acyclic. Hence $H(C) \cong H(P')$.

Shape of P'. The class-c integers in $(-g-N, g)$ are $s_1 - N, s_1, \ldots, s_m$, because $s_1 - N \leq -g < s_1$ and $s_m < g \leq s_m + N$. So P' is the zigzag

$$B_{s_1-N} - A_{s_1} - B_{s_1} - A_{s_2} - \cdots - A_{s_m} - B_{s_m}.$$

Every exponent on it is ≥ 1: $V_{s_j} \geq 1$ since $s_j \leq g-1$, and $H_{s_j} = V_{-s_j} \geq 1$ since $-s_j \leq g-1$. The two end B's each have one A-neighbour. Treat them as edges to a *ground* vertex, at which cochains take the value 0. Then $G \cup \{\text{ground}\}$ is a single cycle with $m+1$ edges.

Levels. Put $x(s) = -\frac{s(s+N)}{2N} - V_s$ and $b(t) = -\frac{t(t+N)}{2N} = -\frac{u^2 - N^2/4}{2N}$, $u = t + \frac{N}{2}$. Then B_s has bottom $x(s) + V_s = b(s)$, and since $H_s = V_s + s$,

$$x(s) + H_s = b(s) + s = \frac{-s^2 - sN + 2sN}{2N} = -\frac{s(s-N)}{2N} = b(s-N),$$

as required for B_{s-N}.

The cokernel part. $\operatorname{coker} \delta_k = H^1(G_k \cup \text{ground}, \text{ground})$, which is $H^1(G_k \cup \text{ground})$ because the ground is one point. This is 0 until the cycle closes, at the maximal peak level M. From then on it is $\mathbb{F}$, and U (restriction of cochains) is an isomorphism. So the cokernel part is a single tower and contributes nothing to τ.

The kernel part. $\ker \delta_k$ is the space of functions on the components of G_k that do not contain the ground. A vertex that is new at level $k+1$ has its edges strictly higher, since all exponents are ≥ 1. So it is isolated at level $k+1$, and the new edges at level $k+1$ join only old vertices or the ground. Hence $\dim(\ker_k / U \cdot \ker_{k+1})$ is the number of merges among old components, including merges into the ground component. That number is $\#(\text{new edges at level } k+1) - \Delta b_1$, and b_1 increases (by 1) only at level M. So $\tau = \{y - 1 : y \in \Pi_c\}$ with one copy of $M-1$ removed, and $|\tau| = (m+1) - 1 = m$. □

5.4 Divisibility

Proposition 5.4. *Suppose $2 \leq |p| \leq 2g-2$ and $S^3_p(K) \cong L \# Y''$ with L an L-space of order $a \geq 2$. Then $|p|$ divides $2g-1$.*

Proof. Write $2g-1 = QN + r$ with $0 \le r < N$. Then $Q \ge 1$, since $N \le 2g-2$. The integers in $[-(g-1), g-1]$ are $2g-1$ consecutive integers, so $n_i = Q+1$ on a set S of r cyclically consecutive residues, and $n_i = Q$ elsewhere. In particular every $n_i \ge 1$.

By Proposition 5.3, $|\tau(HF^+(S^3_p(K), i))|$ equals $n_i - 1$ if $p > 0$ and n_i if $p < 0$; either way it is a strictly increasing function of n_i. By Lemma 5.1 and the invariance of τ up to translation, $|\tau|$ is invariant under $i \mapsto i + b$ with $b \not\equiv 0$. So S is invariant under translation by b.

If $0 < r < N$, then S is a proper nonempty cyclic interval in $\mathbb{Z}/N$. It has exactly one start point (an $x \in S$ with $x - 1 \notin S$), and a translation preserving S must fix that start point. That is impossible for $b \not\equiv 0$. So $r = 0$, i.e. N divides $2g-1$. □

5.5 The key contradiction

Theorem 5.5. *Let K be a nontrivial positive L-space knot of genus g, and p a nonzero integer with $|p| \ge 2$, $|p|$ dividing $2g-1$ and $|p| < 2g-1$. Then $S^3_p(K)$ is not homeomorphic to $L \# Y''$ for any L-space L with $|H_1(L)| \ge 2$ and any closed oriented 3-manifold Y''.*

Proof. Let $N := |p|$ and $n := (2g-1)/N$. Both are odd, $N \ge 3$ and $n \ge 3$. In particular $|p| \le (2g-1)/3 \le 2g-2$, so Proposition 5.3 applies. Suppose such a splitting exists, and let b be as in Lemma 5.1.

(i) The comparison. Let t be the representative of b with $0 < |t| \le (N-1)/2$; it exists because N is odd and $b \not\equiv 0$. By Lemma 5.1 and the invariance of τ:

$$(\star) \qquad \tau \text{ of class } 0 \text{ and } \tau \text{ of class } t \text{ agree up to a translation.}$$

(ii) Arithmetic. For $|c| \le (N-1)/2$, the integers $s \equiv c$ with $|s| \le g-1$ are exactly $c + kN$ for $|k| \le (n-1)/2$. Indeed $|c + kN| \le (N-1)/2 + (n-1)N/2 = g-1$, while $|k| \ge (n+1)/2$ gives $|c + kN| \ge (nN+1)/2 > g-1$. So $m = n$ in every class.

Key fact. Let $u = c + (k + \frac{1}{2})N$ and $u' = c + (k' + \frac{1}{2})N$ with $k \ne k'$. If $u^2 = u'^2$, then $u = -u'$, so $2c + (k + k' + 1)N = 0$. Hence $N \mid 2c$, so $N \mid c$ because N is odd, and $c = 0$. So for $c = t$, distinct k give distinct values of u^2, hence distinct levels in both formulas of Proposition 5.3.

(iii) Case $p > 0$. Class 0: $s_j = (j - \frac{n+1}{2})N$, so the interior peaks have $\mu_j = s_j + N/2 = (j - \frac{n}{2})N$ for $1 \le j \le n-1$. These are $\pm N/2, \pm 3N/2, \ldots, \pm(n-2)N/2$: nonzero $\pm$ pairs. So every element of τ has even multiplicity, and τ is nonempty since $n - 1 \ge 2$. *Class t:* $\mu_j = t + (j - \frac{n+1}{2} + \frac{1}{2})N$, and by (ii) the $n-1$ levels are distinct. A translate of a nonempty multiset in which every multiplicity is even never has all multiplicities equal to 1. This contradicts $(\star)$.

(iv) Case $p < 0$. Class 0: the peaks are $t' = kN$ for $k = -\frac{n+1}{2}, \ldots, \frac{n-1}{2}$, so $u = t' + N/2$ runs over $\pm N/2, \pm 3N/2, \ldots, \pm nN/2$, each value of $|u|$ exactly twice. The maximal level M comes from $|u| = N/2$. Removing one copy of $M - 1$ leaves the levels for $|u| = 3N/2, \ldots, nN/2$ still doubled, and these exist since $n \ge 3$. So τ has a repeated element. *Class t:* by (ii) the $n+1$ peak levels are distinct, so the n elements of τ are distinct. This contradicts $(\star)$. □

6 Proof of the Main Theorem

Assume, as in Section 2, that K is a nontrivial positive L-space knot of genus g, $p \ne 0$, and $Y = S^3_p(K)$ has $k \ge 3$ prime summands.

1. By Lemma 3.1, $Y \cong A \# B$ with B an integer homology sphere, $B \ne S^3$.

2. By Lemma 3.2, $Y \cong L \# Y''$ with L a lens space, hence an L-space, of order $a \ge 2$. So $|p| \ge 2$.

3. K is fibred with fibre of genus g and one boundary component (Theorem E). Proposition 4.1 gives $|p| \leq g$. If $g = 1$, this contradicts $|p| \geq 2$. So $g \geq 2$, and $2 \leq |p| \leq g \leq 2g - 2$.

4. By Proposition 5.4, applied to the splitting of step 2, $|p|$ divides $2g - 1$. Also $|p| \leq g < 2g - 1$.

5. By Theorem 5.5, the splitting of step 2 cannot exist. This is a contradiction.

So $S^3_r(K)$ has at most two prime summands, for every L-space knot K and every slope r. □

7 Remarks

- For a fibred knot, a reducible integral surgery $S^3_p(K)$ with an integer homology sphere summand $B \neq S^3$ requires $|p| \leq g$ (Proposition 4.1). This uses only fibredness.
- Granting Theorem D, the argument of Section 6 shows more: a *reducible* surgery on an L-space knot never has a prime summand that is an integer homology sphere other than S^3, since steps 2–5 use only the homology sphere summand and the lens space summand. (Irreducible surgeries can be homology spheres, e.g. $S^3_{+1}(T(2,3))$ is the Poincaré sphere.) This agrees with the known examples, cables at the cabling slope, whose summands are $L(a,b)$ and $S^3_{b/a}(J)$.

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